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Bihar Engineering University, Patna

B.Tech 1st Semester Exam-2022

Subject: Mathematics-1 (Calculus & Linear Algebra)

Code: 105102

Time: 03 Hours

Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (any seven Question only): **[2×7=14]**

(a) Lagrange's mean value theorem can be proved for a function f(x) by applying Roll's mean value theorem

(i) $\varphi(x) = f(x) + kx^2$
(ii) $\varphi(x) = f(x) - kx^2$
(iii) $\varphi(x) = f(x) + kx$
(iv) $\varphi(x) = \{f(x)\}^2 + kx^2$

(b) The function $f(x) = x(x + 3)\, e^{-x/2}$ satisfy all conditions of Roll's mean value theorem in the interval [-3, 0]. Then the value of c is:

(i) 0
(ii) 1
(iii) 2
(iv) -2

(c) if $\begin{vmatrix} 4/3 \\ 1 \end{vmatrix}$ is an eigenvector of $\begin{vmatrix} 3 & 4 \\ 3 & 1 \end{vmatrix}$. What is the associated eigenvalue?

(i) 4/3
(ii) 5
(iii) -2
(iv) None of the above

(d) if f is continuous on [-2.354, 2.354] then

(i) $\int_{-2.354}^{2.354} f(\cos x)\, dx = 0$
(ii) $\int_{-2.354}^{2.354} f(\cos x)\, dx = 2\int_{0}^{2.354} f(\cos x)\, dx$
(iii) 2.354
(iv) none of the above

(e) let $f(x) = |x|^{3/2}$, $x \text{in} R$ then

(i) f is uniformly continuous
(ii) f is continuous, but not differentiable at $x = 0$
(iii) f is differentiable, and derivative of f is continuous
(iv) f is differentiable, but derivative of x is discontinuous at $x = 0$

(f) if $A(2) = 2i - j + 2k$, $A(3) = 4i - 2j + 3k$, then $\int_2^3 A \cdot dA/dt dt$ is

(i) 5
(ii) 10
(iii) 15
(iv) 20

(g) if $\nabla \times F$. Then it is called.

(i) solenoidal
(ii) Rotational
(iii) irrotational
(iv) None

(h) if $3x + 2y + z = 0$, $x + 4y + z = 0$, $2x + y + 4z = 0$ be a system of equations, then

(i) It is inconsistent
(ii) It has only the trivial solution $x = 0$, $y = 0$, $z = 0$
(iii) Determinant of the matrix of coefficients is zero
(iv) None of these

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(i) If f(x) is continuous in the closed interval [a,b] and f(x) exists in (a,b) and f(a) = f(b), then there exist at least one value $c(a < c < b)$ Such that $f'(c)=0$ is called

(i) Taylor's theorem
(ii) Mac Laurin's theorem
(iii) Rolle's theorem
(iv) Lagrange's mean value theorem

(j) The rank of the matrix $\begin{vmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{vmatrix}$ is

(i) 3
(ii) 2
(iii) 1
(iv) none of these

Q.2 Solve both questions : **[14]**

(a) show that : $\int_0^\infty x^2 e^{-x^4} dx \times \int_\infty^\infty e^{-x^4} dx = [\pi] / [8\sqrt{(2)}]$

(b) Use L' Hospital rule to find the following limits. $\lim_{x \to 0}[\cos - \ln(1+x) - 1 + xy^x] / [\sin^2 x]$

Q.3 Solve both questions : **[14]**

(a) Test the convergence of $1 + 3/7x + 3.6/7.10x^2 + 3.6.9/7.10.13x^3 + [3.6.9.12] / [7.10.13.16]x^4 + \text{ldots}$

(b) Examine the convergence of the series of which the general term is $2^2 4^2 6^2 \text{cdots}[(2n-2)^2] / [3.4.5\text{cdots}(2n-1)2n]x^{2n}$

Q.4 Solve both questions : **[14]**

(a) Obtain the fourth-degree Taylor's polynomial approximation to $f(x) = e^{2x}$ about $x = 0$. Find the maximum error when $0 \le x \le 0.5$

(b) It is given the Rolle's theorem holds the function $f(x) = x^3 + bx^2 + cx$, $1 \le x \le 2$ at the Point $x = 4/3$. Find the value of b and c.

Q.5 Solve both questions : **[14]**

(a) Evaluate $\int_0^\infty \log(x + 1/x)[dx] / [1+x^2]$

(b) Discuss the convergence of the sequence whose n-th term is $a_n = [(-1)^n] / [n] + 1$.

Q.6 Solve both questions : **[14]**

(a) if $f(x) = \log(1 + x)$, $x > 0$ using Maclaurin's theorem, show that for $0 < \theta < 1$, $\log(1 + x) = x - x^2/2 + [x^3] / [3(1+\theta x)^3]$. Deduce that $\log(1+x) =< x - x^2/2 + x^3/3$ for $x > 0$

(b) Using Taylor's theorem, express the polynomial $2x^3 + 7x^2 + x - 6$ in powers of (x - 1).

Q.7 Solve both questions : **[14]**

(a) Find the values of a,b,c if A = $\begin{vmatrix} 0 & 2b & c \\ a & b & -c \\ a & -b & c \end{vmatrix}$ is orthogonal?

(b) Verify Cayley-Hamilton theorem for the matrix A = $\begin{vmatrix} 1 & 4 \\ 2 & 3 \end{vmatrix}$ and find its inverse. Also express $A^5 - 4A^4 - 7A^3 + 11A^2 - A - 10I$ as a linear polynomial in A.

Q.8 Solve both questions : **[14]**

(a) Expand $\ln x$ in power of $(x - 1)$ by Taylor's theorem.

(b) By using Beta function evaluate $\int_0^1 x^5(1 + x^3)^{10} dx$.

Q.9 Solve both questions : **[14]**

(a) Find the Fourier series to represent the function defined as $f(x) = x + \pi/2$, & $-\pi < x < 0$
$\pi/2 - x$, & $0 < x < \pi$

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(b) Evaluate: (i) div F and Curl F, where F= grad (x^3+ y^3+ z^3− 3xyz). (ii) If F = (x + y + z) i + j − (x + y)k show that F·*Curl*F = 0.

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